LITERATURE REVIEW

Representation Learning for Autonomous Peak Detection in Mass Spectrometry Imaging

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Chapter 1

Introduction

The characterization of biological tissues has historically relied upon classical histological staining techniques and light microscopy. However, modern precision medicine might necessitate a significantly deeper understanding of the intricate molecular pathogenesis that underlies morphological changes. To gently bridge this gap, Mass Spectrometry Imaging (MSI), particularly Matrix-Assisted Laser Desorption/Ionization (MALDI), has tentatively emerged as a revolutionary molecular microscopy technique. [Schwamborn and Caprioli 2010] By attempting to map the spatial distribution of thousands of distinct biomolecules directly within an intact tissue section, MSI seems to provide a highly multiplexed view of the cellular microenvironment, which cautiously facilitates critical advancements in biomarker discovery and clinical diagnostics. [Ye et al. 2013]

The theoretical ability to simultaneously capture spatial relationships and complex spectral data without prior knowledge of the target molecules has propelled MSI to the forefront of contemporary biomedical research, facilitating advancements in biomarker discovery, pharmaceutical drug mapping, and clinical diagnostics [Abdelmoula et al. 2021]. In complex localized pathologies such as glioblastoma, colorectal carcinoma, and infectious diseases, MSI enables the direct visualization of tumor heterogeneity, xenobiotic toxins, and localized immune responses that are otherwise entirely invisible to standard hematoxylin and eosin (H&E) staining. Despite these profound advantages and the rapid maturation of the hardware, the clinical translation of MSI into routine diagnostic pipelines is heavily impeded by massive computational bottlenecks. A single high-resolution MSI experiment can generate three-dimensional data hypercubes ranging from tens of gigabytes to several terabytes per tissue slice [Körber et al. 2025]. Each individual pixel within this spatial grid contains a high-dimensional mass spectrum, often consisting of tens to hundreds of thousands of discrete mass-to-charge (m/z) bins [Abdelmoula et al. 2021]. Extracting biologically meaningful, statistically robust signals from this immense, noisy, and highly sparse data structure represents one of the most significant challenges in modern computational biology and medical image analysis [Zhang et al. 2025].

1.1 Problem Statement

Ideally, one might hope that the computational analysis of clinical Mass Spectrometry Imaging datasets would be entirely automated, seamlessly translating high-dimensional raw physical data into robust diagnostic signatures without human intervention. In such a theoretical environment, machine learning models would accurately capture highly non-linear relationships, permitting the unhindered identification of rare, low-abundance molecular biomarkers.

The reality, however, is that current MSI analysis is fundamentally limited by highly heuristic preprocessing pipelines required to compress massive data hypercubes. The most subjective step is the extraction of relevant molecular features through manual algorithmic tuning. Subjective thresholds inevitably discard low-abundance molecular signals, potentially masking crucial biological phenomena permanently, while simultaneously retaining highly structured chemical noise [Lieb et al. 2020].Furthermore, classical filtering might fundamentally distort the underlying non-linear spectral manifold of the data. Navigating away from this bottleneck might require a fundamental reconceptualization of how molecular signals are extracted. [Abdelmoula et al. 2021]

Peak Picking versus Peak Learning

To address specific feedback regarding the fundamental nature of these processes, it seems necessary to explicitly, yet briefly, distinguish between the traditional paradigm of "peak picking" and the emerging paradigm of "peak learning."

The traditional paradigm, commonly referred to as peak picking, is fundamentally a deterministic, rule-based signal filtering task [Li et al. 2022]. Algorithms attempt to separate true molecular ion signals from noise by applying stric geometric and intensity constraints, such as signal-to-noise ratio (SNR) thresholding. In a standard workflow, a human operator is typically required to manually define empirical parameters. The critical limitation here is inherent rigidity and subjectivity. A slight alteration in the SNR threshold can drastically alter the final peak list, making the process highly irreproducible. Perhaps most alarmingly, critical early-stage disease biomarkers often exist in exceptionally low endogenous concentrations; because their signal intensity may fall below the manually defined SNR threshold, peak picking algorithms might blindly erase them from the dataset entirely.[Abdelmoula et al. 2021]

Conversely, the concept of "peak learning" modestly attempts to abandon heuristic thresholds entirely, treating feature extraction instead as an advanced manifold learning and unsupervised representation problem. Utilizing generative probabilistic models, such as a Variational Autoencoder (VAE), the neural network ingests raw mass spectra directly. The network is mathematically forced to compress the massive input data into a tiny abstract latent space and then attempt to reconstruct the original spectrum. To minimize its reconstruction error, the network must autonomously "learn" which specific mass-to-charge (m/z) features represent true biological structures.

Crucially, the "peaks" are extracted not by applying arbitrary thresholds to the raw data, but by mathematically analyzing the trained connection weights of the neural network itself. The nodes corresponding to the most critical molecular ions will naturally possess the largest mathematical connection weights. By cautiously attempting to transition from peak picking to peak learning, researchers hope to gently eliminate human bias and potentially recover subtle, low-abundance biomarkers.

1.2 Expected Contributions

This research modestly aims to contribute a practical, perhaps more accessible computational bridge to the growing field of spatial proteomics. By attempting to mathematically integrate physical neighborhood covariance into an existing, lightweight Variational Autoencoder framework, this project hopes to provide a carefully optimized feature extraction pipeline. It will tentatively evaluate whether we might achieve spatially-aware biochemical mapping and eliminate subjective human bias from MSI preprocessing without taking on the crippling computational and memory burdens typically associated with heavy three-dimensional convolutional networks. If successful, this could offer a more efficient and scalable solution for processing large mass spectrometry imaging datasets in standard laboratory environments.

Chapter 2

Background and Related Work

To adequately contextualize the proposed research, it is helpful to trace the developmental trajectory of computational MSI analysis as documented in recent literature. This progression illustrates a clear transition from classical mathematical heuristics to the current, somewhat complex frontiers of spatial self-supervised deep learning.

2.1 Background: Foundations of MSI

Mass Spectrometry Imaging represents a profound technological intersection between advanced analytical chemistry, physics, and spatial biology. The technology functions by systematically directing a highly focused ionization source across a meticulously defined two-dimensional grid superimposed on a thin biological tissue section. In the MALDI MSI workflow, the tissue is coated with an energy-absorbing organic matrix. A pulsed laser is fired at discrete coordinates, sublimating the matrix and inducing localized desorption and ionization of the endogenous tissue molecules. An analyzer then separates the ions based entirely on their mass-to-charge ratio (m/z).[Caprioli et al. 1997; Körber et al. 2025]

The resulting computational output is an exceptionally high-dimensional data hypercube. For a tissue section imaged at a specific spatial resolution yielding $N \times M$ spatial pixels, each individual pixel contains a complete mass spectrum. This structural complexity creates a profound "curse of dimensionality," where the sheer volume of data rapidly exceeds the capacities of standard analytical workstations. [Abdelmoula et al. 2021; Aspect Analytics 2026]

Due to the immense size, sparsity, and inherent noise of the raw spectral hypercube, classical workflows dictate the application of a heuristic-driven preprocessing pipeline (baseline subtraction, spectral smoothing, intensity normalization, and peak picking) prior to any downstream analysis. Peak picking, the most critical dimensionality reduction step, searches for local maxima that satisfy rigidly defined geometric criteria or SNR thresholds. However, this process relies on strictly linear assumptions. In clinical molecular pathology, critical biomarkers often exist in exceptionally low endogenous abundances. When heuristic SNR thresholds are applied, these vital low-abundance ions are indiscriminately discarded. This severe limitation has driven the field to seek "peak learning" paradigms, wherein the data mathematically dictates feature extraction without arbitrary thresholds. [Abdelmoula et al. 2021; Buchberger et al. 2018]

2.2 Related Work: Deep Learning in MSI

The integration of deep learning neural networks has fundamentally altered the approach to high-dimensional medical image analysis. By leveraging multi-layered neural architectures, deep learning models can automatically extract latent features, accurately model spectrum-structure correlations, and construct low-dimensional spatial representations that preserve critical biological variance while discarding true stochastic noise.

Unsupervised Peak Learning via Generative Probabilistic Models

One of the most significant advancements in addressing the subjectivity of MSI preprocessing is the development of generative probabilistic models. The seminal work by Abdelmoula et al. [2021] introduced "msiPL" (Mass Spectrometry Imaging Peak Learning), a highly sophisticated framework leveraging a fully connected Variational Autoencoder (VAE) to perform unsupervised analysis directly on raw, un-preprocessed MSI datasets.

Unlike conventional algorithms that blindly search for local geometric maxima, msiPL treats the recorded mass spectrum as the output of a complex generative process. The encoder maps the high-dimensional raw spectrum into a lower-dimensional latent space vector, computing the mean and variance parameters that define this distribution. The decoder then attempts to perfectly reconstruct the original spectrum utilizing solely the compressed information.

The paradigm-shifting innovation of msiPL lies in how it executes "peak picking" after network optimization. Rather than applying post-hoc numerical thresholds, msiPL performs backpropagated threshold analysis directly on the learned weight parameters of the neural network. The structural mathematical weights inherently encode the most critical m/z features required for accurate spectral reconstruction. The clinical efficacy of msiPL has been validated on complex datasets (e.g., glioblastoma, colorectal carcinomas), consistently identifying biologically significant m/z peaks that are routinely discarded by classical algorithms. [Abdelmoula et al. 2021 2022]

Spatially Aware Self-Supervised Peak Learning

While msiPL revolutionized spectral analysis, it evaluates mass-to-charge ratios without considering their spatial context across the tissue grid. Biological molecular signals inherently exhibit strong spatial coherence, whereas noise is typically random. To address this, Weigand et al. [2026a] introduced the Spatial Self-Supervised Peak Learning (S3PL) framework.

Building upon foundational autoencoder concepts, S3PL is an end-to-end neural network that performs context-aware peak picking. It employs a 3D convolutional autoencoder that evaluates multi-dimensional spectral patches ($x \in \mathbb{R}^{h \times w \times c}$). The network learns to generate a continuous spatial attention mask based on local spatial coherence, actively penalizing the network's loss function if it attempts to reconstruct non-spatial, stochastic noise. By intrinsically linking peak learning to spatial structure, S3PL isolates highly refined, spatially coherent molecular signals, generating the ideal inputs required for downstream image-based models.

Self-Supervised Spatial Representation and Contrastive Learning

Extracting robust spatial relationships from isolated ion images requires specialized computer vision architectures. Accurately identifying colocalized ions and isotope ions is critical for biomarker discovery. Guo et al. [2024]

introduced "DeepION," a deep learning-based low-dimensional representation model engineered for analyzing mass spectrometry ion images via a self-supervised contrastive learning framework, eliminating the need for manually annotated training data.

DeepION is built upon the SimSiam contrastive model, utilizing twin neural networks with a ResNet18 convolutional backbone. To prevent total model collapse, a prediction multilayer perceptron (MLP) is utilized, and dimensionality reduction via UMAP condenses the representation vector into a 20-dimensional space for efficient spatial similarity calculations.

A critical innovation in DeepION was the introduction of modality-specific data augmentations based strictly on MSI physical prior knowledge. Because ion detection follows Poisson statistical distributions and matrix crystallization causes signal dropouts, the model injects calculated Poisson noise and simulates random missing value patterns during training. The "COL" mode is optimized for colocalized ions, while the "ISO" mode utilizes intensity-dependent missing value operations to discover isotope ions. DeepION achieved remarkable spatial identification accuracy on rat brain tissue, drastically eclipsing standard contrastive models and traditional linear dimensionality reduction techniques.

Deep Learning for Spectral Similarity and Structural Annotation

Once peaks are autonomously extracted and spatially grouped, the final computational hurdle is the structural identification of the molecules those peaks represent. Historically, the "Cosine Score" served as the gold standard for comparing tandem mass spectra against reference libraries. However, cosine similarity suffers from a high false-positive rate and fails to connect structurally similar molecules if minor functional group modifications shift fragment ion masses.

To overcome this, the field has adopted deep representation learning. The Spec2Vec algorithm from Huber et al. [2021b] adapts the Word2Vec NLP architecture to untargeted metabolomics. By training unsupervised on diverse spectra, it maps mass fragments to an abstract, multi-dimensional geometric embedding space. Spectral similarity is calculated based on geometric proximity, demonstrating significantly better correlation with true structural similarity than classical cosine scores.

Building upon this, Huber et al. [2021a]introduces MS2DeepScore which is a highly supervised approach utilizing a deep Siamese neural network. It is mathematically trained to predict the true structural similarity (Tanimoto score) between two molecules directly from their mass spectra. When benchmarking statistical precision across similarity thresholds, MS2DeepScore consistently outperforms both classical cosine metrics and Spec2Vec, allowing for chemically meaningful spatial clustering even when exact library matches are unavailable.

2.3 Summary

Mass Spectrometry Imaging is a transformative technology offering a label-free, detailed window into the biochemical landscapes of intact biological tissues. However, its high-dimensional complexity poses severe computational challenges. Traditional preprocessing pipelines, particularly algorithmic peak picking, dangerously bias biological interpretation by discarding low-abundance signaling molecules and disrupting non-linear spectral manifolds.

The scientific literature demonstrates an accelerating paradigm shift toward deep representation learning architectures to resolve these bottlenecks. Generative models (like msiPL) and spatially-aware extensions (like S3PL) mathematically learn latent spectral manifolds directly from raw data to autonomously identify critical molecular peaks. Self-supervised contrastive networks (like DeepION) utilize physical instrumentation priors to drastically improve the detection of colocalized ions. Finally, structural annotation is now optimized by advanced deep embedding networks, such as Spec2Vec and MS2DeepScore, which model chemical relationships more robustly than classical distance metrics.

Chapter 3

Research Methodology

To carefully address the limitations identified within the literature, the following methodology details the tentative construction, training, and evaluation of the proposed Spatial-msiPL model. The procedural structure emphasizes mathematical rigor and reproducibility while ensuring operations remain constrained to standard public MSI datasets.

3.1 Research Questions

This project is guided by the following specific research questions:

1. How can localized spatial covariance be mathematically integrated into the probabilistic encoding process of a standard 1D Variational Autoencoder without introducing the severe computational and memory overhead associated with 3D convolutional networks?
2. Does the application of an explicit spatial coherence penalty combined with physical noise augmentations effectively isolate true biomolecular signals from stochastic matrix noise?
3. To what extent does this spatially regularized framework improve the identification of biologically relevant, spatially structured peaks compared to standard, spatially-blind models like msiPL?

3.2 Research Aims and Objectives

Aim: To cautiously adapt and evaluate a Spatially-Aware Variational Autoencoder (tentatively named Spatial-msiPL) that might be capable of autonomously identifying spatially coherent molecular peaks in untargeted raw

Mass Spectrometry Imaging data, acting as a potentially computationally efficient alternative to 3D convolutional networks.

Objectives:

- Curate and carefully preprocess high-resolution, raw profile-mode MSI datasets (specifically glioblastoma and colorectal adenocarcinoma models) sourced from public metabolomics repositories. [Abdelmoula et al. 2022]
- Formulate a modified neural network input architecture that ingests contextual vectors, achieved by concatenating a central spatial pixel's spectrum with the averaged spectra of its immediate neighborhood.
- Augment the standard Evidence Lower Bound (ELBO) loss function of a Variational Autoencoder with a targeted spatial penalty term intended to mathematically encourage biological coherence.
- Extract relevant m/z peaks autonomously utilizing a backpropagated mathematical weight analysis, attempting to bypass human-engineered thresholds entirely.
- Quantitatively benchmark the extracted spatial ion images against expert-annotated structural segmentation masks utilizing correlation-based metrics, specifically the mean Spatial F1-score (mSCF1) and Pearson Correlation Coefficients.

3.3 Dataset Selection and Acquisition

To ensure strict scientific reproducibility, and per the guidance to utilize independently sourced data, the proposed model will be developed and evaluated exclusively against diverse, high-resolution public datasets. These will be sourced from the ProteomeXchange consortium and the National Metabolomics Data Repository (NMDR). Additionally, if supplementary data is required, raw imzML datasets can also be acquired from the METASPACE knowledge base. A strict methodological requirement for this investigation is the exclusive use of raw, profile-mode mass spectrometry data stored in the standard imzML format. As previously noted, profile-mode data gently preserves the non-linear morphological wave-shapes and minor baseline artifacts that are entirely stripped from centroided data. This rich mathematical complexity is arguably necessary for representation learning models to successfully differentiate signal from noise.

Specific benchmark datasets cautiously selected for this study include:

1. **Glioblastoma Multiforme (GBM) Dataset (PR001292):** Acquired via a highly sensitive 9.4 Tesla SolariX FT-ICR mass spectrometer, this dataset maps tissue from an intracranial patient-derived xenograft (PDX) mouse model of glioblastoma. The extreme high-mass resolution and the highly heterogeneous nature of the tumor boundaries make this an ideal stress test for spatial peak learning models seeking subtle, low-abundance ions. This is the dataset originally utilized in the Abdelmoula et al. [2022] studies.
2. **Human Colorectal Adenocarcinoma (CAC) Dataset (MTBLS415):** Available from MetaboLights, providing an alternate pathological landscape to ensure model generalizability beyond neurological tissues.

Crucially, expert-annotated structural segmentation masks corresponding to these specific tissues will be acquired from the open-source repositories associated with the S3PL benchmark to serve as the objective ground truth for evaluation. [Weigand et al. 2026a]

3.4 Preprocessing Pipeline

Although the primary goal of the model is to eliminate heuristic dimensionality reduction, basic data scaling is mathematically required to stabilize deep learning gradients during training.

- **Total-Ion-Count (TIC) Normalization:** Each raw profile spectrum will undergo TIC normalization to ensure uniform intensity scaling across the tissue sections, modestly mitigating ionization efficiency artifacts that naturally occur during MALDI acquisition [Deininger et al. 2011].
- **Contextual Input Re-engineering:** To carefully circumvent the need for computationally expensive 3D convolutions while still attempting to introduce spatial awareness, the data input space will be fundamentally restructured. For any given pixel coordinate (x, y), the algorithm will mathematically isolate the spectrum of that central pixel, denoted as $x^{(i)}$. It will then compute the mean spectrum of its immediate 8-pixel Moore neighborhood [Winderbaum et al. 2015]. A new "contextual vector", $\tilde{x}^{(i)}$, will be generated by concatenating the central spectrum with the spatially averaged neighbor spectrum.

- **Physical Prior Injections:** Inspired by the apparent success of the DeepION architecture [Guo et al. 2024], synthetic Poisson statistical noise will be dynamically injected into the input arrays during the training phase. This cautiously simulates the physical ion counting statistics inherent to mass spectrometer detectors, theoretically preventing the network from merely memorizing localized matrix crystallization artifacts.

3.5 Models and Algorithms Used: Spatial-msiPL

The core architecture of Spatial-msiPL is proposed to build upon a standard, fully connected Variational Autoencoder, utilizing multi-layer perceptrons (MLPs) [Abdelmoula et al. 2021] rather than the heavy convolutional blocks seen in S3PL [Weigand et al. 2026a].

The network fundamentally consists of an Encoder network, $q_\phi(z|\tilde{x})$, and a Decoder network, $p_\theta(x|z)$. The probabilistic Encoder will ingest the high-dimensional contextual vector $\tilde{x}^{(i)}$ and attempt to compute the mean (μ) and variance (σ^2) parameters, defining a Gaussian distribution in a highly compressed latent space z [Abdelmoula et al. 2021]. Crucially, the Decoder network will be programmed to attempt to reconstruct *only* the original, isolated central spectrum $x^{(i)}$, rather than the entire contextual input. It is hoped that this deliberate architectural bottleneck will force the model to utilize the broader spatial neighborhood context to effectively "denoise" the localized output spectrum.

3.5.1 Loss Function and Spatial Coherence Penalty

A standard VAE is optimized by maximizing the Evidence Lower Bound (ELBO), which carefully balances the reconstruction fidelity against the Kullback-Leibler (KL) divergence [Abdelmoula et al. 2021]. To mathematically enforce biological structure, this project tentatively proposes augmenting the objective function with a novel, explicit spatial regularization term, $\lambda\mathcal{L}_{spatial}$.

This explicit penalty will continuously calculate the Euclidean distance between the generated latent representation of a central pixel and the mean latent representations of its adjacent physical neighbors.

The proposed total loss function might be formally expressed as:

$$\mathcal{L}_{total} = \mathbb{E}_{q_\phi(z|\tilde{x})}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|\tilde{x})||p(z)) + \lambda \sum_{j\in\mathcal{N}(i)} ||\mu_\phi(\tilde{x}^{(i)}) - \mu_\phi(\tilde{x}^{(j)})||^2$$

Where $\mathcal{N}(i)$ represents the immediate spatial neighborhood of pixel i, and λ acts as a tunable hyperparameter dictating the strength of the spatial smoothing. By attempting to minimize this specific loss configuration, the network might actively discourage extreme latent space divergence between adjacent pixels, potentially acting as a powerful filter against stochastic, spatially random mass spectrometry noise [Winderbaum et al. 2015].

3.5.2 Unsupervised Peak Extraction via Backpropagated Weight Analysis

The hoped-for elimination of the classical "peak picking" step relies heavily on mathematical feature extraction post-training. Following successful convergence and stabilization of the loss function, the network weights will be frozen [Abdelmoula et al. 2021].

Instead of generating arbitrary local maxima thresholds, the Spatial-msiPL model will attempt to execute a backpropagated analysis directly on the learned connection weights $W^{(1)}$ linking the input contextual vector layer to the first hidden layer of the neural network [Abdelmoula et al. 2021]. In a fully connected network, each input node corresponds directly to a specific m/z spectral bin. Through the mechanics of backpropagation via gradient descent, the network has iteratively updated these weights ($\Delta W = -\eta \frac{\partial \mathcal{L}}{\partial W}$) over thousands of epochs in an attempt to minimize reconstruction error. Consequently, it is assumed that input nodes with large resulting weight values are mathematically identified by the network as the most critical features required for accurate, spatially coherent spectral reconstruction.

The relative importance of each specific mass-to-charge bin (m/z_k) will be objectively determined by calculating the $L2$-norm across all hidden units (H) for that specific input node [Abdelmoula et al. 2021]:

$$Importance(m/z_k) = \sqrt{\sum_{h=1}^{H} (W_{h,k}^{(1)})^2}$$

The m/z values demonstrating the highest structural weight importance will be subsequently extracted and carefully cross-referenced to the nearest local maximum on the average spectrum to form the final, threshold-free "learned" peak list [Abdelmoula et al. 2021].

3.6 Evaluation Method and Statistical Validation

A pervasive difficulty in untargeted mass spectrometry imaging informatics is the apparent absence of an absolute, mathematically objective biological "ground truth" across tens of thousands of potential ions. To cautiously circumvent reliance on subjective visual inspection, this research plans to implement a rigorous, spatially structured quantitative evaluation protocol utilizing the previously acquired expert-annotated histological segmentation masks [Weigand et al. 2026a].

3.6.1 The Mean Spatial F1-Score (mSCF1)

The primary evaluation metric will be the mean Spatial Correlation F1-score (mSCF1), recently introduced in the literature specifically for benchmarking spatial fidelity in MSI datasets [Weigand et al. 2026a].

The proposed evaluation pipeline will cautiously proceed as follows:

1. **Ion Image Generation:** For every m/z peak tentatively identified by the Spatial-msiPL model, the 2D spatial ion distribution image will be plotted across the tissue coordinates.
2. **Pearson Correlation Coefficient (PCC):** The 2D spatial PCC will be carefully calculated between the extracted spatial ion image and the binary, expert-annotated structural segmentation mask provided in the benchmark dataset.
3. **Thresholded Binary Classification:** To evaluate whether the model successfully isolated structured biological signals from stochastic noise, the continuous PCC values will be thresholded. Following established protocols, four distinct empirical correlation thresholds will be utilized: $T_{PCC} \in \{0.3, 0.4, 0.5, 0.6\}$ [Weigand et al. 2026a].
4. If an extracted ion image exhibits a PCC greater than or equal to T_{PCC} against *any* annotated biological structure in the mask, it will be classified as a True Positive (implying a biologically meaningful spatial peak).

5. Conversely, if the ion image fails to meet the threshold for any structure, it will be categorized as non-informative stochastic noise, or a False Positive.
6. **mSCF1 Calculation:** For each of the four thresholds, standard Precision, Recall, and F1-scores will be calculated based on the identified peaks. The final reported metric will be the unweighted mean of these four F1-scores, yielding the mSCF1 [Weigand et al. 2026a].

This stringent methodology is intended to ensure the proposed architecture is evaluated strictly on its capacity to uncover meaningful spatial biological architecture, rather than its ability to merely isolate high-intensity mass spectrometry matrix clusters.

3.6.2 Baseline Comparisons

To rigorously contextualize any potential performance gains of the proposed Spatial-msiPL architecture, it will be computationally benchmarked against several established methods:

- **Classical Baselines:** Standard heuristic peak picking utilizing the R-based MALDIquant framework will provide a baseline for traditional methods [Gibb and Strimmer 2012].
- **1D Baselines:** The original, spatially blind msiPL Variational Autoencoder will be tested, ensuring an exact assessment of any performance gain yielded specifically by the added spatial penalty [Abdelmoula et al. 2021].
- **3D Baselines:** If hardware permits, the heavy 3D-convolutional S3PL network might be utilized to measure computational runtime efficiency and memory consumption against the proposed 1D model [Weigand et al. 2026a].

Table 2 outlines the planned comparative metrics:

- **mSCF1 Score:** Assess spatial fidelity of extracted peaks. Target Baseline Comparison: msiPL (1D) and MALDIquant.
- **PCC Distribution:** Analyze correlation strength with tumor masks. Target Baseline Comparison: S3PL (3D Convolutional).
- **GPU Memory Usage:** Quantify hardware requirements during training. Target Baseline Comparison: S3PL (3D Convolutional).

- **Inference Runtime:** Measure processing speed for clinical scalability. Target Baseline Comparison: msiPL and S3PL.

Chapter 4

Work Schedule

The project is tentatively structured to be executed over a 4-month academic period, logically divided into the following iterative phases to ensure continuous progression and allow time for troubleshooting.

- **Phase 1 (Literature & Data):** The primary focus will be to consolidate the literature review, navigate the NMDR and ProteomeXchange repositories, and gently acquire and parse the multi-gigabyte raw `imzML` profile datasets (PR001292 GBM and MTBLS415 CAC) and their corresponding segmentation masks. Managing the sheer size of these files may present early challenges that need to be carefully mitigated.
- **Phase 2 (Preprocessing & Baseline):** The next step will involve implementing the basic data preprocessing pipeline, including TIC normalization and the generation of the contextual vectors. An attempt will also be made to replicate the baseline msiPL model locally to ensure a fair comparative analysis later.
- **Phase 3 (Model Development):** This phase will be dedicated to architecting the Spatial-msiPL network itself. The spatial coherence penalty will be integrated into the ELBO loss function, and training runs will commence with the Poisson noise augmentations. It is anticipated that hyperparameter tuning, specifically for the λ penalty, will require patience and multiple iterations.
- **Phase 4 (Evaluation & Documentation):** The final month will focus on extracting the final peak lists via backpropagation analysis, executing the comparative benchmarking using the mSCF1 and PCC metrics, and carefully compiling the results into the final Honours research report.

Chapter 5

Conclusion

The computational bottleneck caused by heuristic peak picking currently appears to remain one of the primary barriers preventing the seamless integration of Mass Spectrometry Imaging into automated, clinical diagnostic workflows. While early unsupervised spectral generative models, such as the foundational 1D msiPL framework, have successfully suggested the viability of autonomous mathematical feature extraction, their failure to utilize structural biological context might inherently limit their diagnostic accuracy and noise-filtering capabilities. Conversely, 3D convolutional models seem to provide high spatial fidelity but impose somewhat restrictive computational burdens that could limit their clinical scalability in standard laboratory settings.

The tentatively proposed Spatial-msiPL architecture seeks to modestly bridge this critical gap. By cautiously attempting to integrate physical neighborhood covariance directly into the encoding process via contextual input vectors and an explicit spatial coherence penalty, it is hypothesized that spatial awareness might be achieved within a highly efficient 1D framework. By rigorously evaluating this architecture using spatial correlation metrics against expert-annotated tumor masks, this preliminary research hopes to contribute a potentially more accessible deep learning framework capable of autonomously mapping the biochemical topography of diseased tissues. If successful, this gentle approach might offer a computationally modest step toward reducing human bias in spatial omics analysis.

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Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ✓
2. Sourcing Related Work (summarising, identifying sources) ✓
3. Methods and Experiment Design (experiment setup, model tuning) ☐
4. Data Analysis (presentation, coding, interpretation) ☐
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☐
7. Presentation (rendering graphics, formatting) ✓
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If other uses were involved, please specify below:

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If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

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